

Simulation of the Fluid Flow-Related Phenomena in the Electrolyte of an Aluminum Electrolysis Cell

YUFENG WANG, LIFENG ZHANG, and XIANGJUN ZUO

A full-scale water model and mathematical simulation were used to study the fluid flow-related phenomena in a water model of an aluminum electrolysis cell. The time-dependent, multiphase fluid flow model was developed to represent the complex transient flow in the electrolysis bath. The accuracy of the mathematical model was validated by the ink dispersion and laser doppler velocimetry measurements in the water model. The shape, motion, release frequency of large-size bubbles, the fluid flow pattern, and the electrolyte-metal interface were predicted by the mathematical simulation. The design and operation of the anode were discussed, including the effect of the anode edge corner shape, the presence of a tilted bottom angle, and the magnitude of applied current density. The results indicated that coupling using a curved corner, with slot and with tilted angle at the anode, is effective for the release of bubbles and for the stability of the electrolyte-metal interface.

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I. INTRODUCTION

THE Hall-Héroult process is the main industrial process for the production of primary aluminum. Figure 1 illustrates a schematic of this process. Alumina powders are dissolved in the molten cryolite (electrolyte). A direct current is applied and passes through the entire aluminum electrolysis cell. Because of the electrochemical reaction, the liquid aluminum is deposited at the bottom of the bath, whereas the carbon monoxide (CO) and carbon dioxide (CO₂) form at the bottom of the anode. The presence of these nonconductive gases under the anode contributes to the overall resistance between the anode and the cathode.^[1,2] Bubbles cover from 50 pct to as much as 90 pct of the anode bottom area,^[1,3,4] which causes the average voltage drop in the range of 0.15 to 0.35 V.^[4,5] The situation could be much worse, like the occurrence of anode effect, if a gas film formed under the anode.^[6,7] Stable and efficient operation of the electrolysis process involves the removal of anode gas bubbles^[8] and the oscillation of the electrolyte-metal interface.^[9] Key factors include the following^[10,11]:

- (a) Behavior of bubble: nucleation, growth, motion, interaction, coalescence and detachment from the anode

- (b) Design of anode: material, dimension, shape, slot geometry, and edge curvature
- (c) Bath chemistry: dissolution and concentration
- (d) Operation parameters: applied voltage, current density, anode cathode distance, and anode-tilted angle

To reduce the energy consumption and stabilize the electrolysis bath, the study on the fluid flow-related phenomena and the behavior of bubbles is extremely important. To investigate electrolysis bath, the industrial measurements can provide information such as the current variation,^[12] electrolyte-metal interface fluctuation,^[13] and bath temperature.^[14]

However, because of the harsh environment and high-temperature condition, it is impossible to measure the gas film under the anode. Laboratory-scale aluminum electrolytic cells have been built to evaluate the anode overvoltage and the distribution and release frequency of bubbles^[15–17] as well as to quantify the volume of gas.^[1] An aluminum electrolysis cell was established to visualize the bubbles underneath the anode using ultrasound technology.^[4]

Water modeling is another approach to study the multiphase phenomena in the reduction cell. Fortin^[18] studied the effect of a slope anode on the behavior of bubbles in 1984 using water modeling. Compared with the electrolysis experiment, the hydrodynamic model is cheap to operate and safe for measurement. Because of the advantage of transparent properties of water, the bubbles could be visualized by a camera and the velocity of the fluid flow can be measured by laser doppler velocimetry (LDV) or particle image velocimetry. Since 1990, water modeling was widely applied to study the bubble behaviors under the anode.^[8,19–22] A series of studies based on water modeling was carried out by Shekhar and Evans.^[23–25] The design of a cell, including the interelectrode gap and anode groove, was discussed.

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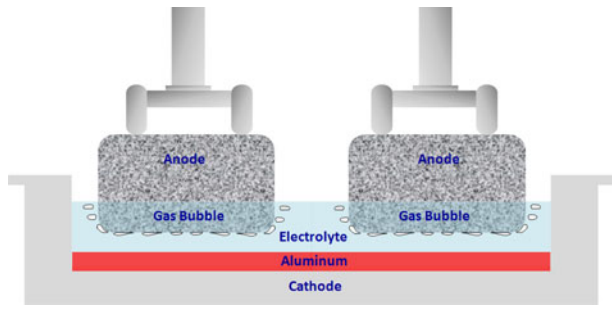


Fig. 1—Phenomena in aluminum electrolysis cell.

Recently, water modeling was employed to study the deformations of the electrolyte–aluminum interface, which is close to the side channel of the cell.^[26]

With high-speed computers, a numerical model can predict the complicated flow pattern in aluminum reduction cells.^[9,17] Literatures have reported that the mass transport and heat-transfer processes in the electrolysis cell are dominated by the magneto-hydrodynamic (MHD),^[17,27,28] gas-driven,^[29,30] and thermal gradient driven forces.^[14] However, in the electrolyte bath, the gas-driven force is more dominant than others, and the motion and release of bubbles from the anode dominate the mixing and convective transport in the electrolyte bath according to Bilek^[31] and Doheim.^[32] Many studies only focus on the electrolyte–gas system.^[26,33] In the electrolyte–gas system, the fact that bubbles release from the anode is a transient or pseudoperiodical phenomenon.^[4] Two-dimensional (2D) transient numerical models have been used to simulate the gas-release process and the transient bath–metal interface has been predicted.^[26] However, the 2D model cannot simulate the effect of the anode slot to track the release frequency of bubbles at the different positions inside the anode bath. Therefore, three-dimensional (3D) models are desired to solve this issue in the electrolyte–gas system.

To the best of the authors' knowledge, 3D simulation is rarely reported to reveal the releasing process of bubbles under the anode cell. In the present study, a 3D transient, multiphase numerical model was developed to study the flow-related phenomena in a water model of the electrolyte bath of a reduction cell. Only the gas–water system is considered in the current study. The effect of other driven forces will be considered future studies. The simulation results were validated by water modeling measurements. The mathematical prediction was combined with water modeling results, and the effects of the corner shape of the anode, tilted angle and slot on the motion of bubbles, release frequency, fluid flow pattern, and the stability of the electrolyte–metal interface were analyzed.

II. METHODOLOGY

A. Water Modeling

Table I lists the properties of water and cryolite. Because of the similar kinematic viscosity between water and the cryolite,^[18] the water–air system can be used to

Table I. Properties of Water and Cryolite^[18,23,24]

Liquids	Cryolite	Water
Temperature (K [°C])	1223 (950)	298 (25)
Density (kg/m ³)	2.1	1.0
Kinematic viscosity (m ² /s)	1.4×10^{-6}	1.0×10^{-6}
Surface tension (N/m)	0.5	0.073

model the aluminum electrolysis process (cryolite and CO/CO₂ bubbles). Figure 2 shows the schematic of the water model. The model was created to represent one of the anodes in a modern smelter. A Plexiglas tank was used as a container. An anode with a slot at the bottom was modeled by suspending two Plexiglas boxes in the tank that were 1500-mm long, 170-mm wide, and 430-mm high. The distances between the anode and the bottom tank, the anode-tilted angle, and the slot dimension are adjustable. The main purpose of the water model was to study the effect of the detaching bubble. The coalescence is the main growth mechanism of bubbles under low-temperature condition and the nucleation of bubbles on the anode has little effect on the motion of larger bubbles. In contrast, the small bubbles under the anode surface do not significantly affect the motion of the large bubbles.^[34] Several smaller bubbles between the large bubbles will be swallowed by the larger bubbles along the anode.^[1] According to the observation of Fortin,^[18] gas behavior was dominated by the motion of large bubbles across the anode surface. Therefore, the bubble formation and the motion of small bubbles were disregarded in the current study. Gas bubbles under the anode were simulated by injecting air through 16 plastic pipes, each with a diameter of 4 mm.

Assuming the gas generated at the anode is CO₂, one molecule of CO₂ could be released because of the passage of four electrons through the circuit. Therefore, the correlation between the current density and gas flow rate is expressed as follows^[35]:

$$q = \frac{i \times R \times T}{4F \times P} \quad [1]$$

where q is the gas generation rate at the unit anode bottom surface, m³ s^{−1} m^{−2}, i is the anodic current density, A m^{−2}, R is the universal gas constant, 8.314 J mol^{−1} K^{−1}, T is the absolute temperature, 1223 K (950 °C) for electrolyte, F is Faraday's constant, 96,486 A mol^{−1} s^{−1}, and P is atmosphere pressure, 101,325 Pa. These data give

$$q = 2.6 \times 10^{-7} i$$

The flow rate of the gas generated (in m³/s) can be determined by multiplying q with the area of the anode bottom surfaces, which is $1.5 \times 0.17 \times 2 = 0.51$ m² in the current study. The gas flow rate used in the water modeling by Fortin *et al.*^[18] and Shekhar *et al.*^[8,23–25] satisfied Eq. [1], as shown in Figure 2(c). According to Eq. [1], the gas flow rate in the current water model was set within 16 to 157 L/min, which is in the range of 0.5 to 5 m³/s per m², or 0.05 to 0.5 cm³/s per cm² and is equivalent to a 0.2 to 2.0 A/cm² current density, as

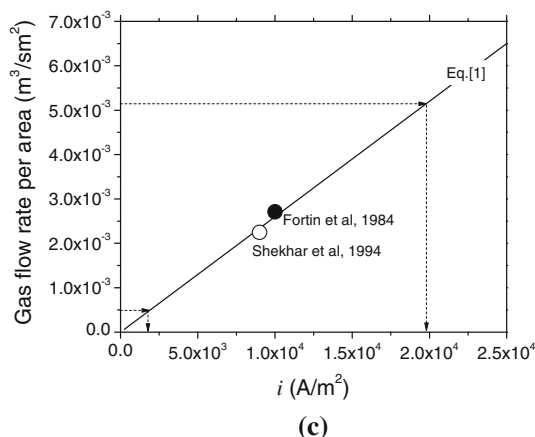
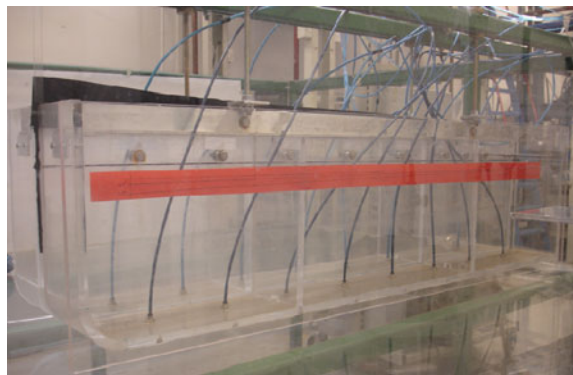
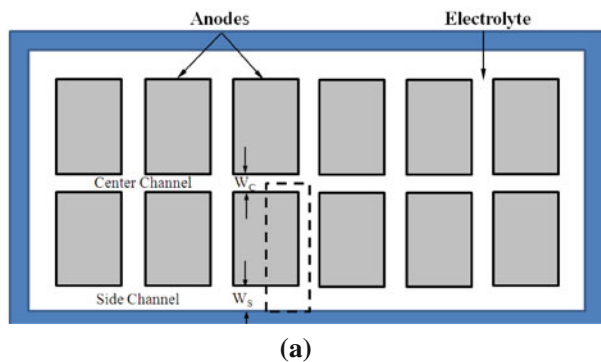


Fig. 2—Schematic of the water model. (a) Schematic top view. (b) Real image of water model. (c) Gas flow rate per area generated with the current density (Fortin *et al.*^[18] and Shekhar *et al.*^[23,25]).

shown in Figure 2(c). The gas flow rate roughly generates a similar gas fraction in the water modeling to that in the aluminum reduction cell where the carbon anode is consumed to give off approximately 2.5 m³ of CO₂ for every kilogram of aluminum produced.^[19] The detailed parameters used in water modeling are listed in Table II.

A video recorder was used measure the geometry and trajectories of bubbles and the bubble coverage area under the anode bottom. To visualize the flow pattern, blue ink was used as a tracer. The fluid flow in the water model was characterized using a two-component LDV, by which the local instantaneous velocity can be measured at some points. For each point, the mean velocity can be obtained by averaging the measurement

Table II. Parameters Used in the Water Modeling

Parameters	Value
Depth of electrolysis bath (ACD)	40 mm
Anode slope	0 deg, 0.4 deg, 1.1 deg
Width of slot	3 mm
Width of side channel, W _s	390 mm
Width of center channel, W _c	107 mm
Anode length, width, and height	1500 mm, 340 mm, 430 mm
Gas flow rate	16 to 157 L/min, or 0.52×10^{-3} to 5.13×10^{-3} m ³ /s m ²

of 1000 particles over approximately 5 to 8 minutes. The fluid flow features at the following different locations were measured: the region of side channel and the region under the anode, as shown in Figure 3. Because the LDV measurement is from point to point, more than 300 points were measured at the side channel region, and more than 100 points were counted at the region under the anode. The turbulent energy per unit mass k (m²/s²) was calculated as follows from the LDV measurements including the components of horizontal velocity $\bar{u}$ (m/s) and vertical velocity $\bar{v}$ (m/s)^[36]:

$$k = \frac{1}{2} [\bar{u}^2 + \bar{v}^2] \quad [2]$$

It should be realized that the bubble size has a significant effect on the flow pattern. The bubbles formed in electrolysis should be smaller than those in the current water modeling. To simulate the formation of bubbles under an anode by water modeling, a porous plate might be a good choice, such as in the studies by Shekhar and Evans^[23,24] and Kiss.^[37,38] However, the main interest of the current work is to study the effect of detaching bubble sheets. Therefore, the effect of bubble size is neglected. It was assumed that bubble sheets were formed prior to the anode edge. Gas evolution was simulated by injecting air through a microporous high-density polyethylene plate in Fortin's study.^[18] It indicated that small bubbles were created on the entire anode surface. These small bubbles moved slowly. As they began to coalesce, they gained a velocity. Then larger bubbles traveled along the anode surface face, overtaking and swallowing many small bubbles. Bubble growth was rapid and continuous until it was released at the anode edge. The study concluded that, on a horizontal anode, the process of bubble nucleation, growth, coalescing, and release involved little bubble motion and led to a gas film. On the inclined surface, the gas behavior was dominated by the motion of large bubbles across the anode surface. In the study of Thonstad and Ngoya,^[34] small bubbles forming on the anode surface and had little effect on the motion of the large bubbles. In the study of Aaberg,^[1] it was proven that the gas layer at the anode was dominated by large, slab-like bubbles. Several smaller bubbles are likely present between the slabs, possibly residing at the anode

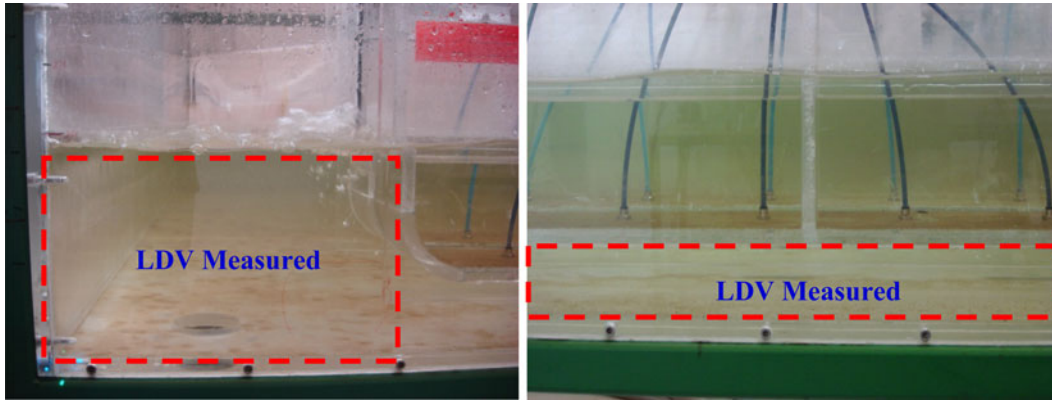


Fig. 3—LDV measured region.

during their formation and initial growth. Such small bubbles will be “swallowed” by the larger bubbles sweeping along the anode. As far as the authors know, the similar water model setup has been applied to study gas behavior in the studies of Zoric *et al.*^[2] and Einarsrud *et al.*^[26]

B. Mathematical Models

The multiphase transient turbulent fluid flow in the electrolysis bath was simulated by calculating the continuity equation and Navier–Stokes equations, using the standard $k - \varepsilon$, two-equation turbulent model.

The equation of continuity is expressed as follows:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_i)}{\partial x_i} = 0 \quad [3]$$

where ρ is density (kg/m^3), t is the time (seconds), u_j is velocity component in x_j direction (m/s), and x_j is direction along different axis (m).

The Navier–Stokes equation is expressed as follows:

$$\rho \frac{\partial(\rho u_i)}{\partial t} + \rho \frac{\partial(u_i u_j)}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu_{\text{eff}} \frac{\partial u_i}{\partial x_j} \right) + \frac{\partial}{\partial x_j} \left(\mu_{\text{eff}} \frac{\partial u_j}{\partial x_i} \right) + \rho g + \vec{F} \quad [4]$$

where p is pressure (pa), $\vec{F}$ is a momentum source/sink term (N) used to incorporate the electromagnetic force,^[39] and μ_{eff} is the turbulence-adjusted effective viscosity (kg/m s).

Turbulent kinetic energy and its dissipation rate ($k - \varepsilon$ equations) are expressed as follows:

$$\frac{\partial(\rho k)}{\partial t} + \frac{\partial}{\partial x_i} \left(\rho u_i k - \frac{\mu_{\text{eff}}}{\sigma_k} \frac{\partial k}{\partial x_i} \right) = \mu_t \frac{\partial u_j}{\partial x_i} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \rho \varepsilon \quad [5]$$

$$\begin{aligned} \frac{\partial(\rho \varepsilon)}{\partial t} + \frac{\partial}{\partial x_i} \left(\rho u_i \varepsilon - \frac{\mu_{\text{eff}}}{\sigma_\varepsilon} \frac{\partial \varepsilon}{\partial x_i} \right) \\ = C_1 \mu_t \frac{\varepsilon}{k} \frac{\partial u_j}{\partial x_i} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - C_2 \frac{\varepsilon}{k} \rho \varepsilon \end{aligned} \quad [6]$$

With the $k - \varepsilon$ equations, the turbulent viscosity μ_t (kg/m s) is expressed as follows^[40,41]:

$$\mu_t = \rho C_\mu \frac{k^2}{\varepsilon} \quad [7]$$

The volume of fluid (VOF) multiphase model was employed to track the free surface moving through the computational grid by simultaneously solving another parameter—the volume of fluid per unit volume f_i . It requires the addition of the following conservation equation^[42]:

$$\frac{\partial f_i}{\partial t} + \left[\frac{\partial}{\partial x_i} (f_i u_i) \right] = 0 \quad [8]$$

The VOF method was used to solve the fluid flow problems. It is an efficiency-solving algorithm for free surface tracking. This approach has a powerful volume-tracking feature. The method is based on a fraction function f in Eq. [8]. It is defined as the integral of a fluid’s characteristic function in a computational grid cell. Basically, when the cell is empty (no traced fluid value), the value of f is zero. If the cell is full, the value of f is 1, and when the interface cuts the cell, f is a number between 0 and 1. During the calculation, the fluid volume fraction in each grid cell is consistent with the resolution of the other flow quantities. If the amount of fluid in each cell is known, then it is possible to locate the interface, as well as determine the interface slope and curvatures. The interface can be located because they lie in cells partially filled with fluid or between cells full of fluid and cells that have no fluid. Because of its simplicity and robustness, VOF has been widely used in multiphase simulation. The VOF scheme is limited for non-Newtonian behavior and requires finer mesh, which may require a long computation time, especially for 3D calculations.

Both 2D and 3D simulations were performed in the current study. The structural grid was designed to simplify the calculation domain. The total number of grid points was 485,000 and 660,000 for the 2D and 3D models, respectively. The high-quality structural grid system ensures the high efficiency and accurate prediction. During the calculation, an unsteady Navier–Stokes solver coupled with a VOF model were used to simulate

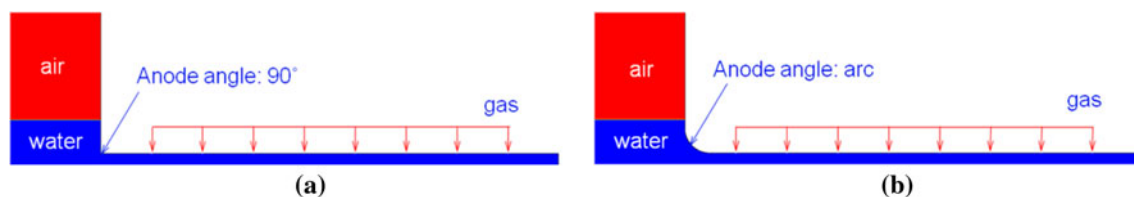


Fig. 4—Geometry and boundaries of the 2D model. (a) Anode edge, 90 deg shape. (b) Anode edge, curved shape.

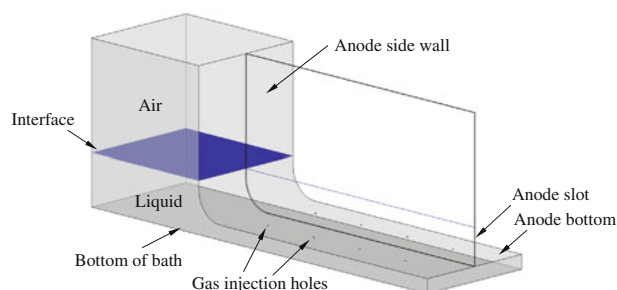


Fig. 5—Geometry and boundaries of the 3D model.

Table III. Parameters and Boundary Conditions for 3D Model

Parameters	Value	Parameters	Value
Anode slope	0 deg, 1.1 deg	Inlet k value	$0.0001 \text{ m}^2/\text{s}^2$
Width of slot	3 mm	Inlet ε value	$0.0001 \text{ m}^2/\text{s}^3$
Width of side channel	390 mm	Liquid density	998.2 kg/m^3
Inlet diameter	4 mm	Gas density	1.225 kg/m^3
Inlet velocity	5.32 m/s	Surface tension	0.073 N/m

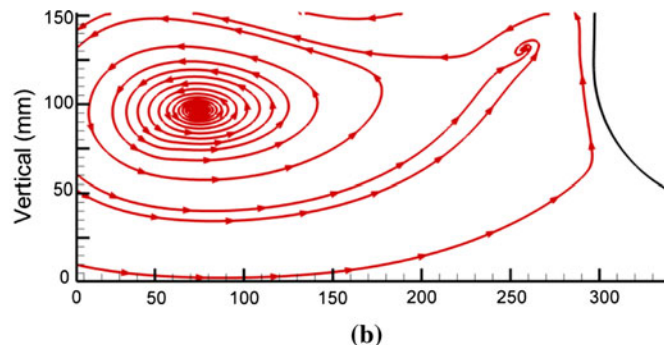
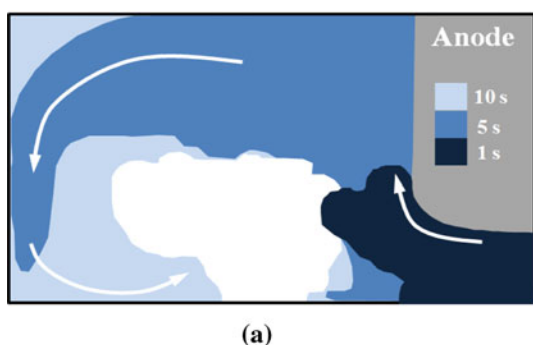


Fig. 6—Recirculation flow at the side channel. (a) Tracer dispersion in the water model. (b) Flow pattern stimulated by 3D model (at 10 s).

air and water flow. The entire domain was assigned to both fluids (water and air) that are discretized using a one-fluid formulation with changing physical properties (viscosity and density) across the interface. A conservative approach for Navier–Stokes equations is used in order to well maintain the mass balance of each phase. The commercial software, FLUENT, is used to perform the current calculations.

The model domain is illustrated in Figures 4 and 5. The gas inlet velocity was set from 0.5 to 1.0 m/s according to the current density; the pressure outlet at the top surface was set for the gas release. A 2D model is feasible to study the effects of anode-tilted angle and the corner curvature shape of the anode, but it is impossible to study the effect of the slot. The setting for the anode-tilted angle was either 0 deg or 1.1 deg, and the design for the anode corner was 90 deg or a curved shape (with 45 deg round off). A 3D model was also used to study the effect of the anode slot. The combination effect of the anode-tilted angle and slot also was investigated. The parameters used in the 3D simulation are listed in Table III.

III. MODEL VALIDATION

Figure 6 shows the comparison of the flow pattern at the side channel in water modeling and in numerical simulation. In the water model, the ink was injected into the tank approximately 500 mm away from the anode edge. The numerical simulation was time dependent, and the flow pattern in the side channel was assumed stable after the gas injection at 10 seconds. Both models were conducted with a tilted anode. The gas flow rate was $0.2 \text{ cm}^3/\text{s}$ per cm^2 . The center plane (the cross section on the column of gas injection pipes) was selected for validation. The numerical simulation shows a similar recirculation flow at the side channel. The accuracy of numerical simulation was also verified by the LDV measurement. The LDV measurement and simulation conditions were the same as the previous validation. As shown in Figure 7, both the distribution and the magnitude of the velocity vector from the measurement and simulation were approximately identical. The comparison proves the accuracy of the current numerical simulation.

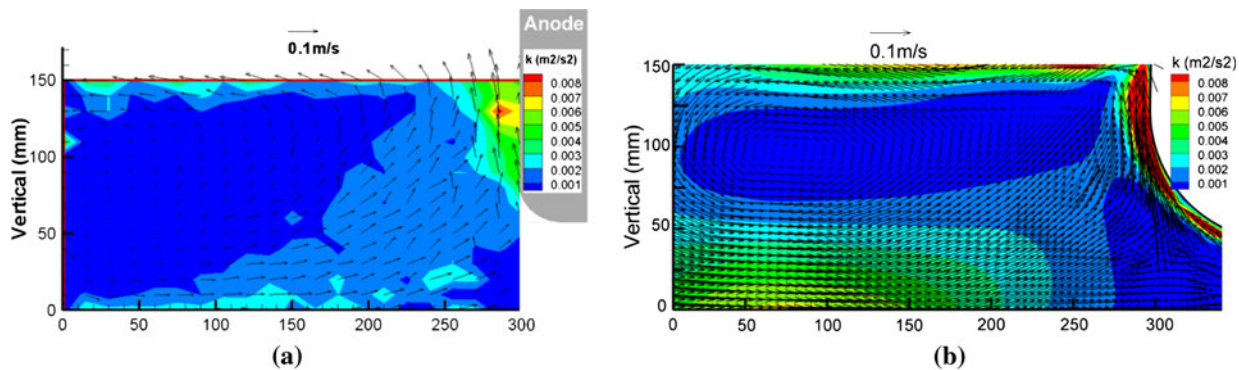


Fig. 7—Distribution of velocity vector. (a) Measured by LDV. (b) Stimulated by 3D model (at 10 s).

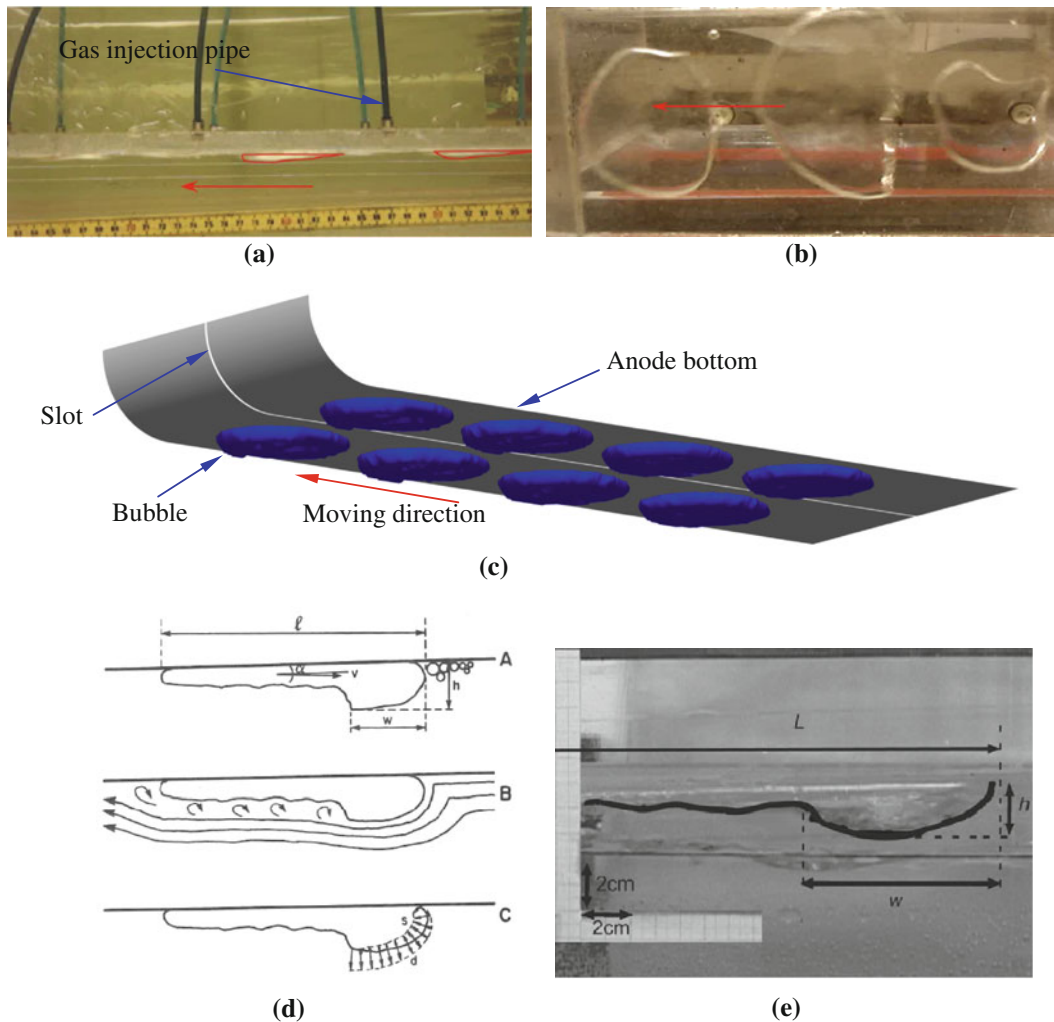


Fig. 8—Shape of bubbles. (a) Water model side view. (b) Water model bottom view. (c) 3D simulation. (d) Longitudinal dimension of the bubbles observed in Fortin's study.^[18] (e) A typical view of a bubble travelling under the anode in the water model of Einarsrud.^[26]

IV. SHAPE OF BUBBLES

The shape of bubbles under the anode has a thick bubble front and a thin, long trail portion, as shown in Figure 8. In the bubbles observed from the anode bottom, the largest dimension is transversal to the motion direction. When the tilted angle is large enough,

big gas pockets are formed by coalescence by sweeping away smaller bubbles. The shape of these large gas pockets is not circular anymore, having nearly straight leading and trailing edges. The thickness of the bubble-laden layer is generally determined by the maximum possible height of a bubble under the anode. The big gas

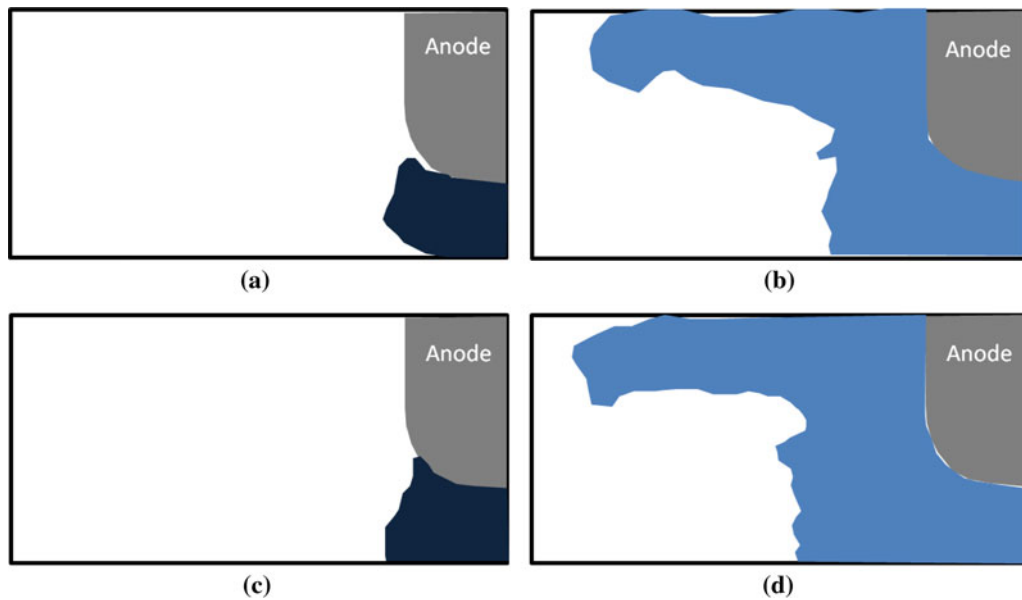


Fig. 9—Tracer dispersion (anode-tilted angle 1.1 deg). (a) Gas flow rate $0.13 \text{ cm}^3/\text{s}$ per cm^2 , $t = 0 \text{ s}$. (b) Gas flow rate $0.13 \text{ cm}^3/\text{s}$ per cm^2 , $t = 6 \text{ s}$. (c) Gas flow rate $0.26 \text{ cm}^3/\text{s}$ per cm^2 , $t = 0 \text{ s}$. (d) Gas flow rate $0.26 \text{ cm}^3/\text{s}$ per cm^2 , $t = 4 \text{ s}$.

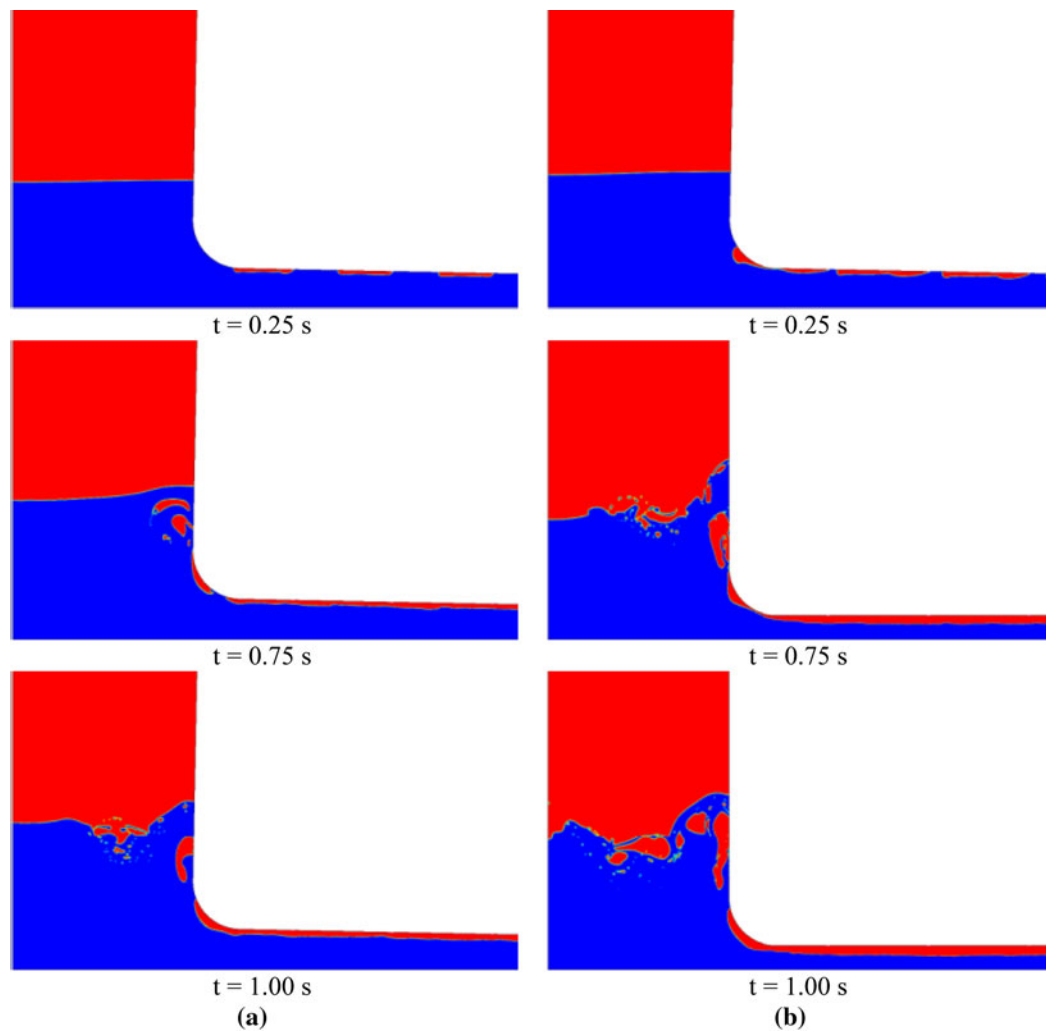


Fig. 10—Effect of gas flow rate on bubble motion and free surface emulsification. (a) Gas inlet velocity: 0.5 m/s . (b) Gas inlet velocity: 1.0 m/s .

pockets have a “head” at the leading edge, which is about twice as thick as the overall thickness of the bubble layer. One important parameter is the bubble coverage fraction on the anode bottom. The coverage fraction is larger than 80 pct for both the mathematical model and water model observation. The large gas pockets were also observed and discussed in other studies,^[18,26,34] and is shown in Figures 8(d) and (e). Fortin *et al.* reported the longitudinal dimension of bubbles (Figure 8(d))^[18] that was measured parallel to the direction of bubble motion at the center of the anode. The “length” varied from 2 to 128 cm. The transverse dimension of the bubbles varied from 4 to 40 cm where the entire width of the anode slice is 40 cm.

V. EFFECT OF GAS FLOW RATE AND ANODE SHAPE

The gas flow rate is proportional to the electrolysis current density. The gas release frequency increases with an increasing gas flow rate. The gas flow rates were set as $0.13 \text{ cm}^3/\text{s}$ per cm^2 and $0.26 \text{ cm}^3/\text{s}$ per cm^2 to represent the current densities of $0.5 \text{ A}/\text{cm}^2$ and $1.0 \text{ A}/\text{cm}^2$, respectively. The ink was injected from the third gas pipe from the corner of the anode. The time starts from 0 seconds when the ink arrives to the corner of the anode. Based on the observation of ink dispersion under the anode, it took around 6 seconds for the tracer to move from the anode bottom to the side when the gas flow rate is $0.13 \text{ cm}^3/\text{s}$ per cm^2 , whereas, it took less time

for the tracer to arrive at the same position, approximately 4 seconds, as illustrated in Figure 9.

Figure 10 shows the 2D simulation of the nucleation and motion of bubbles. The large flow rate, corresponding to a large current density, could accelerate the flow in the bath and induce the instability close to the air–fluid interface. The larger flow rate induces more vigorous emulsification of the interface between the gas and the liquid.

For the real reduction cells, the rate of gas formation is governed primarily by the current density. However, the shape of the anode dominates the gas flow conditions at any given flow rate.^[43] The electrolysis experiments have indicated that anode shape affected the

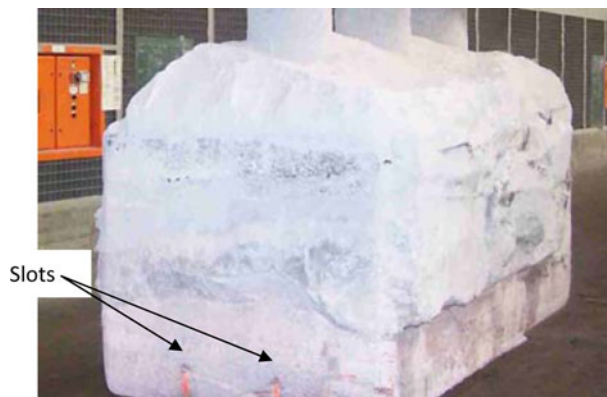


Fig. 12—Slot configuration.

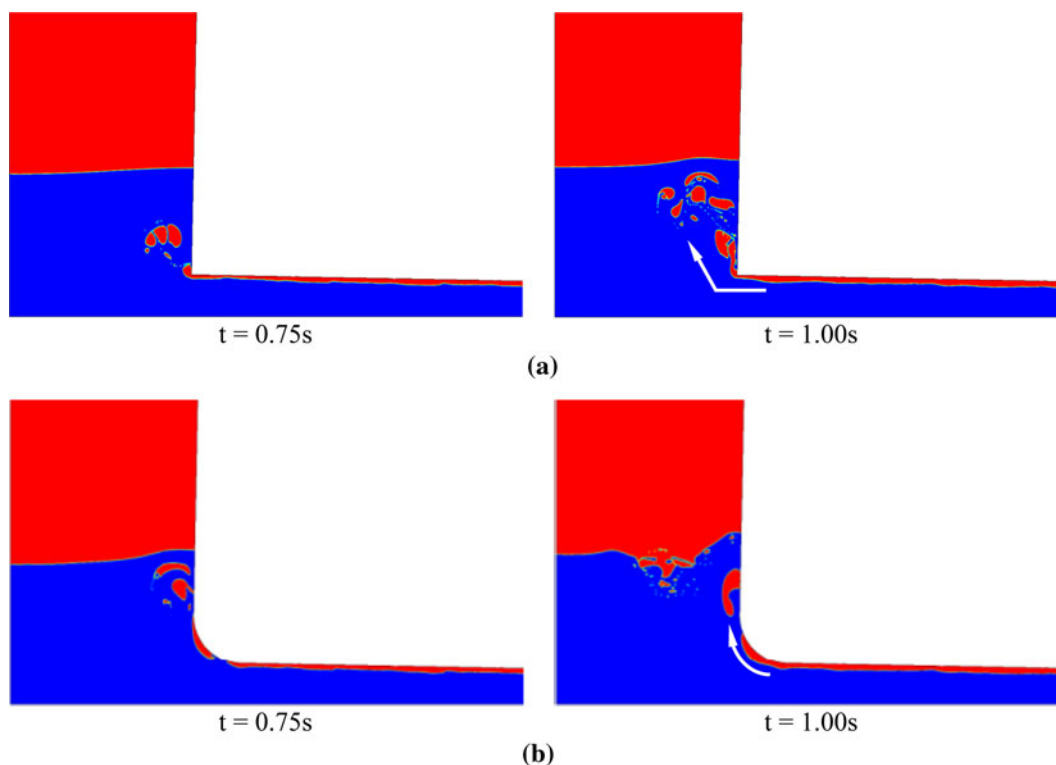


Fig. 11—Effect of anode shape on bubble motion. (a) Anode corner, 90 deg shape. (b) Anode corner, curved shape.

release frequency and size distribution of bubbles and the cell resistance.^[16] In the current study, a 2D simulation was used to study the effect of anode corner shape. The inlet gas velocity was fixed at 1.0 m/s. The following anode corner shapes were discussed: one with a 90 deg turning edge and the other with a curved anode corner. The curved corner anode is a mature anode or is designed intentionally. The design of the anode corner shape was discussed in this section. As shown in Figure 11, the main difference between the two anode corner shapes was the detachment point of the bubble. For the former, because of the horizontal velocity, the bubble kept moving for a certain distance, and then floated up and formed a bubble plume near the center of the side channel. The 90 deg anode corner design delays the floatation of bubbles. For the latter, the bubble could move along the curved anode edge. The detached point was high and close to the anode, which allowed for bubbles to float up to the bath surface. Therefore, the curved anode edge is more efficient for the bubble detachment and escaping than the 90 deg anode corner design.

VI. EFFECTS OF TILTED ANGLE AND SLOT DESIGN OF THE ANODE

The effect of anode-tilted angle on the fluid flow of a reduction cell was studied by a water model as early as 1984.^[18] The role of the anode slope was studied experimentally,^[20,44] and the effect of the anode-tilted angle (from 2 deg to 30 deg) on the volume and the terminal velocity of bubbles were discussed. Recently, the effect of a 1 deg slope anode was simulated by Kiss *et al.*^[37]

Another import design for the anode is anode slot, as shown in Figure 12. The idea of “grooved” anodes was introduced and its effect on the fluid flow in the electrolysis cell was investigated by a water model in 1990.^[8] The anode slot is effective for modifying the formation, release, and coverage of bubbles under an anode.^[1,3] Splitting one anode into two by a slot was attempted in the industrial reduction process at Hydro Aluminum (Oslo, Norway) in 1993.^[45] The anodes were first implemented in a Rio Tinto Aluminum (London, UK) managed plant in 1998.^[12] Today, the anode slot

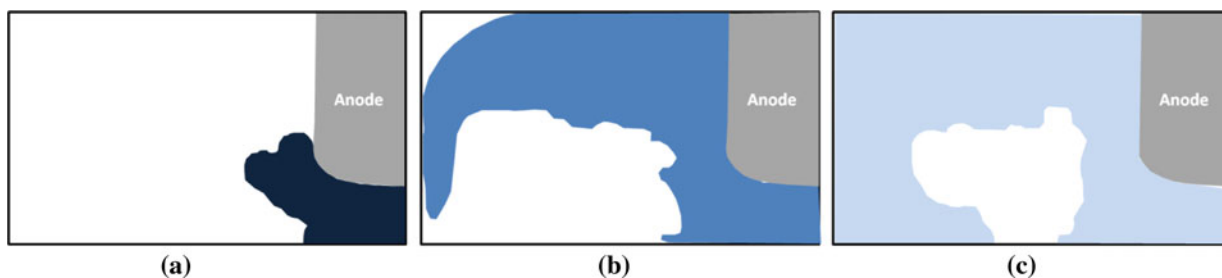


Fig. 13—Tracer dispersion at different times. (a) Time = 1 s. (b) Time = 5 s. (c) Time = 10 s.

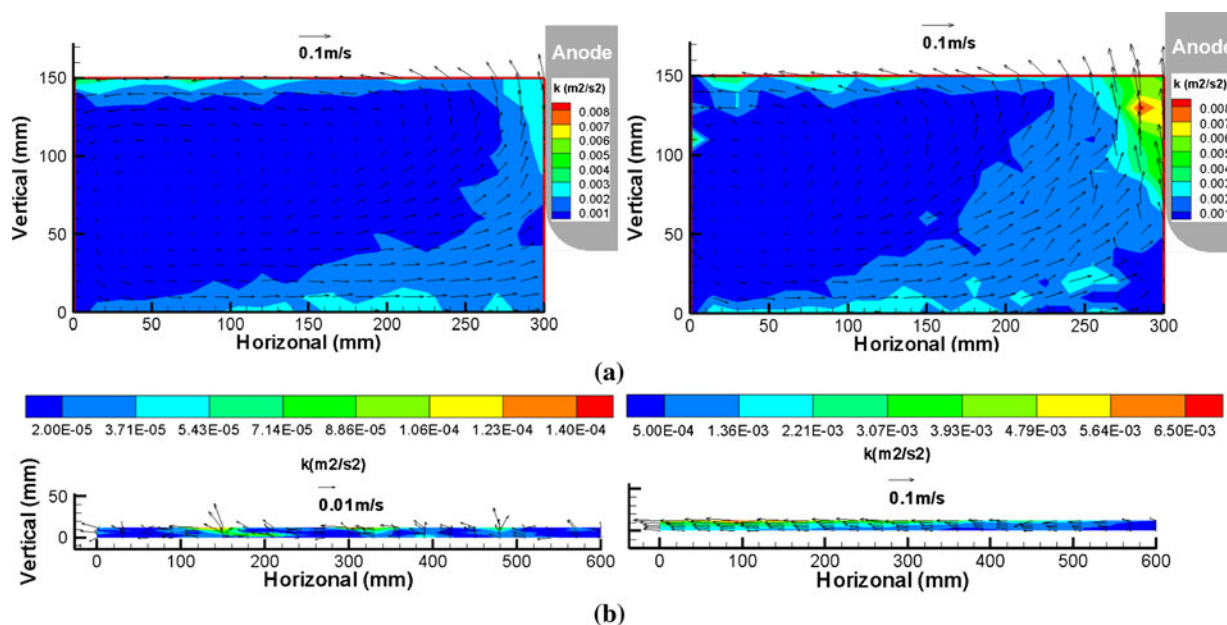


Fig. 14—Measured velocity and turbulent energy (left: anode-tilted angle, 0 deg, gas flow rate, 0.2 cm³/s per cm². Right: anode-tilted angle, 1.1 deg, gas flow rate, 0.2 cm³/s per cm²). (a) Side channel. (b) Under anode.

design has been widely accepted in primary aluminum industries. It was reported that a 30 pct gas layer under the anode was reduced^[46] and \$14.00 per day could be saved for each electrolysis cell with the presence of slots in the anode.^[47] In the current study, both water modeling and a mathematical simulation were applied to study the combining effect of anode-tilted angle and anode slot.

Because the curved anode corner design has been approved for bubble releasing and floating, all of the following modeling work considered the curved anode corner. The gas flow rate was $0.2 \text{ cm}^3/\text{s}$ per cm^2 , and the anode-tilted angle varied from 0 deg to 1.1 deg. A slot with a width of 3 mm is at the center of the anode. According to the flow pattern indicated by ink dispersion, with a 0 deg tilted angle, the tracer stagnated under the anode for a long time. With the tilted angle, the ink tracer quickly entered the side channel and generated a recirculation flow, as shown in Figure 13. With larger

tilted angle, the ink entered the side channel more quickly and generated a stronger recirculation. The continuously floating bubbles is the reason for the recirculation flow.

According to the LDV measurement, the high-turbulence region was located at the gas plume region in the side channel and close to the free surface, which was mainly dominated by the momentum transfer from rising bubbles to the liquid and energy dissipation of the waves emulsification induced by the escaping of the bubbles. The center plane with a gas injection pipe was measured. With a 0 deg tilted angle, the large velocity points were randomly scattered under the anode because bubbles did not easily escape from the end of the anode, as shown in Figure 14; and the gas film expanded under the anode and rose through the side wall. With a 1.1 deg tilted angle, gas bubbles moved smoothly toward the end of the anode, and the velocity distribution under the anode was more uniform than with the 0 deg tilted

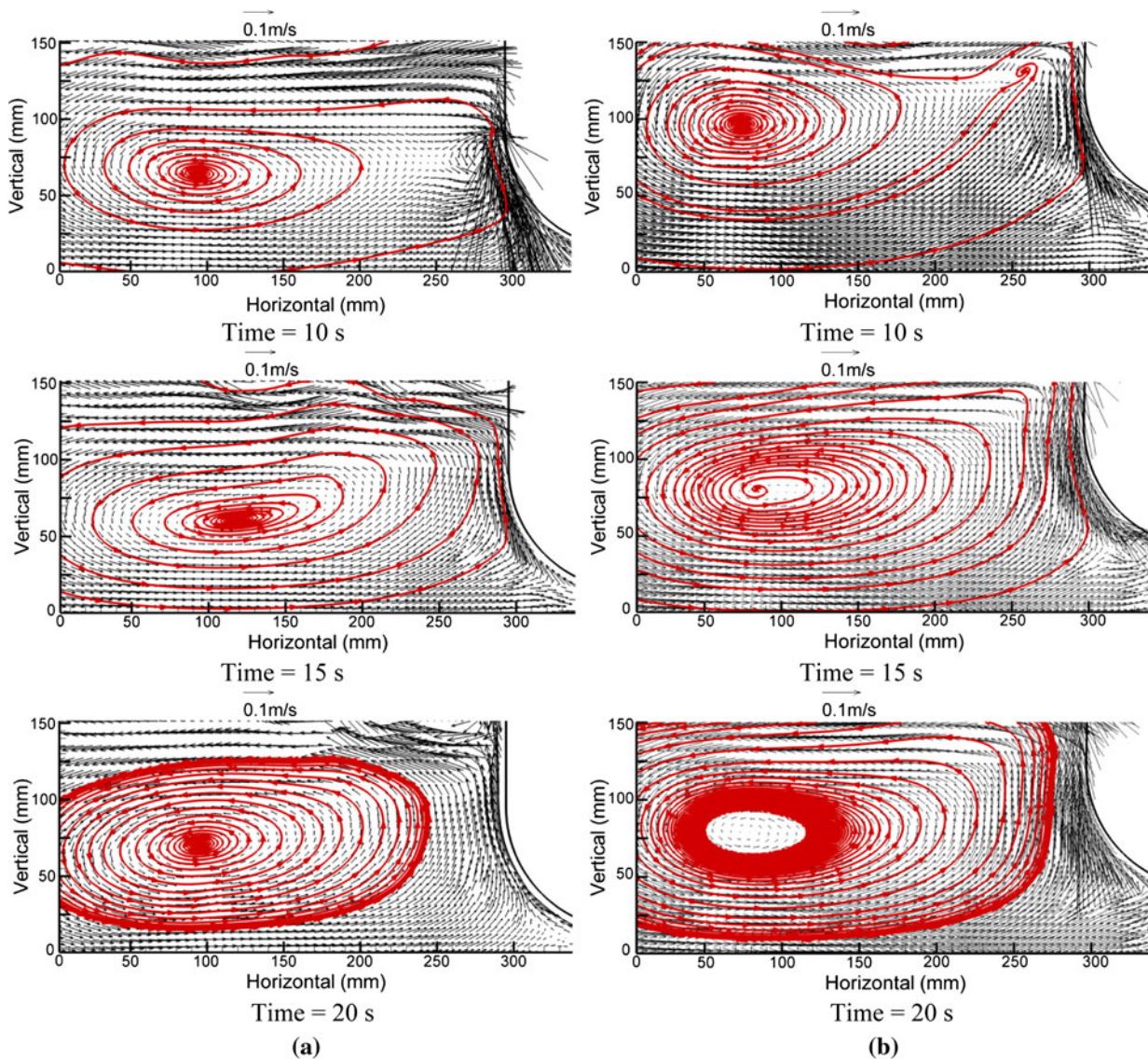


Fig. 15—Calculated 3D velocity and streamline on the anode center plane of the side channel. (a) Anode-tilted angle, 0 deg. (b) Anode-tilted angle, 1.1 deg.

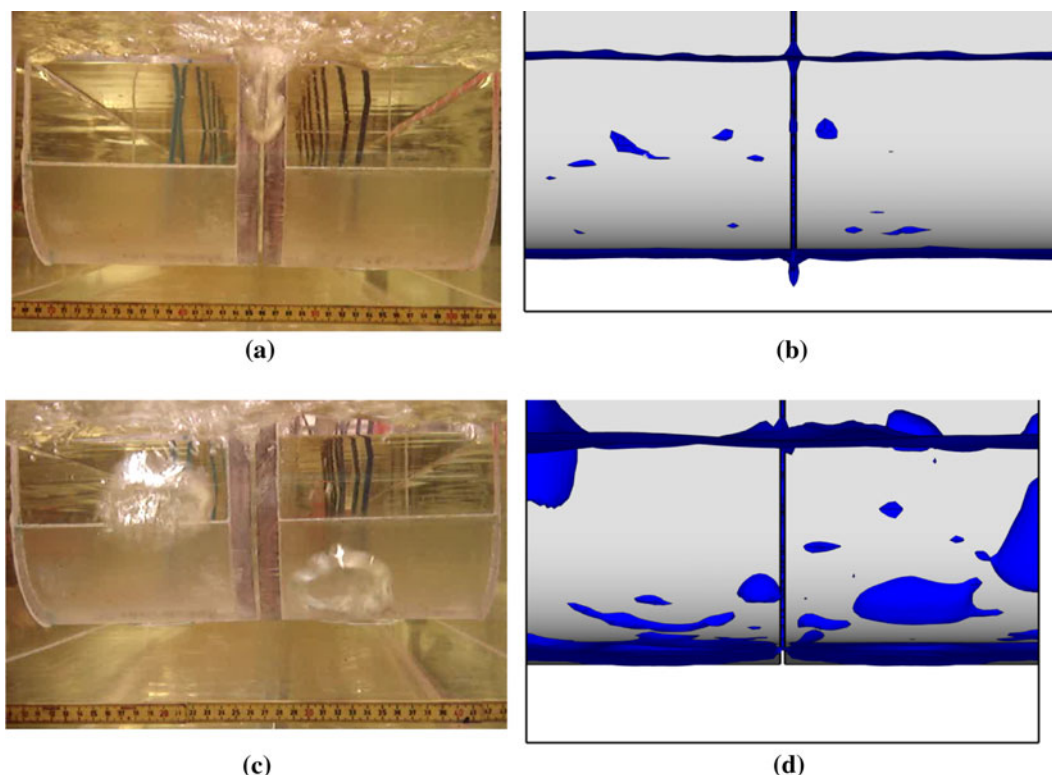


Fig. 16—Bubbles released from anode bottom. (a) Water model observation (0 deg tilted angle). (b) 3D simulation (0 deg tilted angle). (c) Water model observation (1.1 deg tilted angle). (d) 3D simulation (1.1 deg tilted angle).

angle. A larger circulation flow pattern at the side channel was also generated because of the presence of the gas plume.

The recirculation flow in the channel also was proven by the numerical simulation. Figure 15 shows the results of the 3D transient simulation. The flow patterns shown here were from 10 to 20 seconds after gas injection. The pseudoperiod releasing bubbles induced the fluctuation inside the electrolysis bath. With the larger tilted angle anode, the recirculation flow was stronger.

For the release path of bubbles, many bubbles escaped from the slot and occasionally from the curved edge of the anode with a 0 deg tilted angle, whereas, bubbles escaped from both the slot and the anode edge if the tilted angle at the anode was 1.1 deg, as shown in Figure 16. Also, compared with the tilted anode, the coverage fraction of the gas film under the nontilted flat anode is much larger, as indicated in Figure 17 (approximately 90 pct). From the previous analysis, the presence of the slot is useful to allow for bubble release and escape. It provides an alternative for the gas bubbles to escape. The combination of the slot and the tilted angle of the anode were more effective.

VII. BUBBLE RELEASE FREQUENCY

As one of the important parameters for aluminum electrolysis process, the bubble release frequency is generally measured by counting the voltage or current fluctuation from industrial data. According to the

literature,^[4,18] bubbles released from an anode during aluminum electrolysis occur at a frequency of 0.5 to 3 Hz under the normal anode current density. A typical measurement was illustrated in Figure 18.

Because a time-dependent transient model was developed in the present study, the velocity changing with time could be monitored. The variation of velocity inside the aluminum electrolysis bath is a good indicator for bubble release. The release of a large bubble induces a large fluctuation, whereas the small bubbles have a minor effect on the bath velocity fluctuation. Therefore, the magnitude of velocity fluctuation would reflect the bubble size, and the variation of velocity would indicate the bubble release frequency. In the current study, two points near the gas–liquid interface in the water model were monitored in the 3D simulation: point 1 is in the front of the anode and point 2 is in the front of slot, as illustrated in Figure 19. The 0 deg anode-tilted angle needed 4 to 8 seconds for the bubble release to reach a steady frequency, whereas it only needed 3 seconds for the 1.1 deg anode-tilted angle. The bubble release frequency was around 1 Hz for the 0 deg tilted-angle anode, and was approximately 1.5 Hz for the 1.1 deg tilted-angle anode. The simulated bubble release frequency is pseudoperiodical, which is identical to the industrial measurement.^[4] For a given current density, the lower the bubble release frequency, the greater the bubble size under the anode^[16]; therefore, the larger bubbles are under the flat anode, which was confirmed by the water model observation. Moreover, for the 0 deg tilted anode, the bubbles had difficulty moving

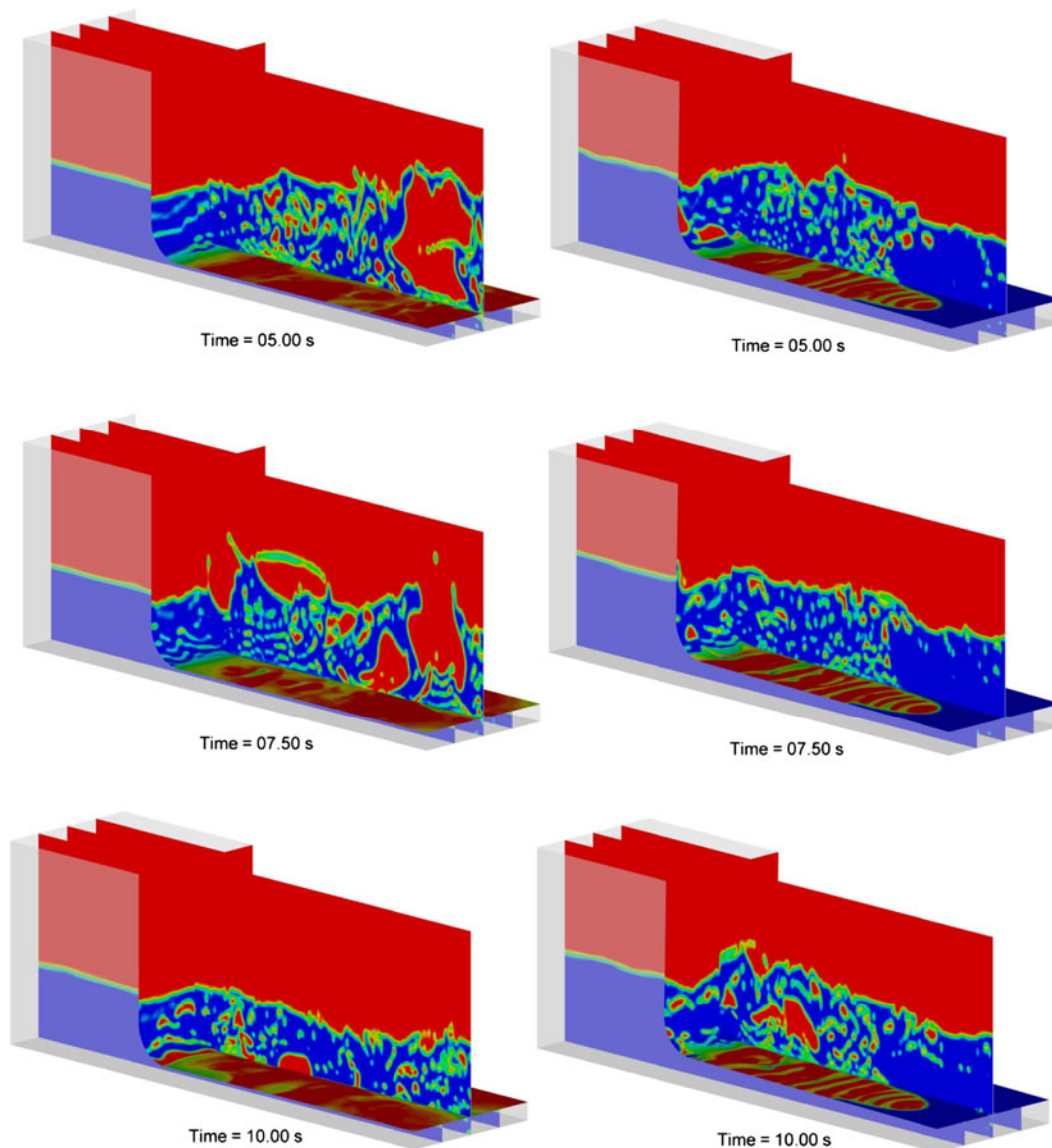


Fig. 17—Bubbles released from anode bottom.

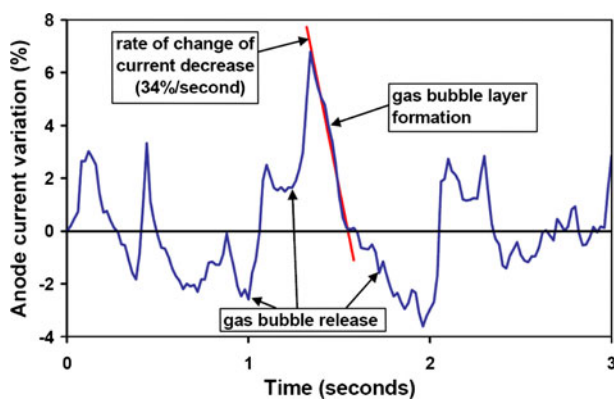


Fig. 18—Typical anode current variation from unslotted anode showing bubble release events in a real industrial cell (sample at 50 Hz).

forward under the anode. With the continuous increasing gas volume, the bubbles could touch the edge of the anode slot, then be broken and float up. The velocity fluctuation at point 2 was much larger than at point 1 in Figure 19(a), indicating that the bubbles mainly escaped at the slot and that few bubbles were released at the anode edge. However, in Figure 19(b), the velocity fluctuations at points 1 and 2 are almost the same, indicating that the anode slot and the 1.1 deg tilted anode had an identical effect on bubble release and escape.

VIII. PRESSURE AT THE INTERFACE BETWEEN THE ELECTROLYTE AND MOLTEN ALUMINUM

Both gas-driven and MHD-driven flow cause oscillations in the electrolyte–metal interface. These oscillations

could affect the primary aluminum production process; for example, the turbulence close to the interface induced by oscillations could influence the heat transfer in the reduction cell.^[9] A general electrolyte–metal profile was measured by a special inductive probe.^[13] Also a mathematical model was applied to predict the interface. In a study by Fraser,^[9] a variety of interface shapes were simulated by considering the effect of MHD. In the work of Torstein *et al.*,^[29] the pressure created from gas-driven flow was considered an important factor in the deformation of the electrolyte–metal interface, and a relative height difference on the interface was predicted as follows:

$$\Delta h = \frac{p_{\text{local}} - p_{\text{mean}}}{g(\rho_{\text{metal}} - \rho_{\text{bath}})} \quad [9]$$

where Δh is the relative height difference of electrolyte/metal interface; p_{local} is the local pressure at the interface; p_{mean} is the average pressure at the interface;

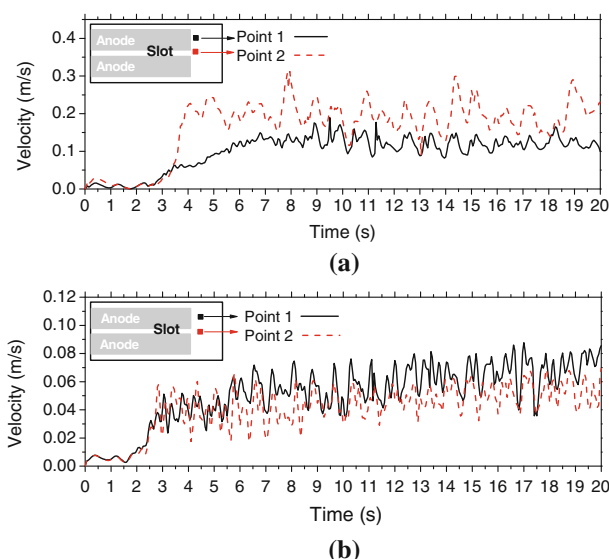


Fig. 19—Bubble release frequency. (a) Anode-tilted angle, 0 deg. (b) Anode-tilted angle, 1.1 deg.

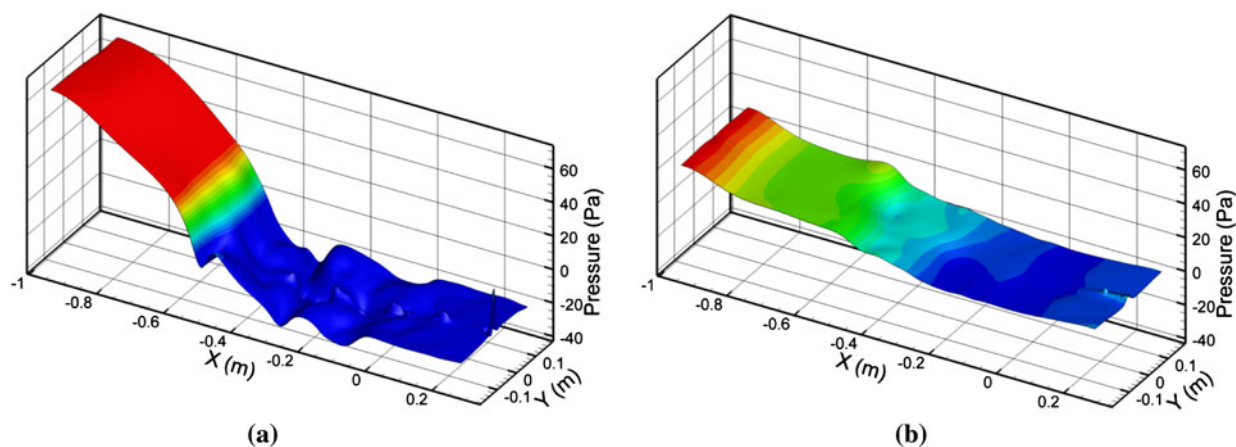


Fig. 20—Metal/electrolyte profile predicted by pressure difference at 10 s after gas injection. (a) Anode-tilted angle, 0 deg. (b) Anode-tilted angle, 1.1 deg.

g is gravity acceleration; and ρ_{metal} and ρ_{bath} are the densities of the aluminum metal and the electrolyte bath, respectively.

Because the relative height difference is proportional to the pressure difference, the pressure difference on the electrolyte–metal profile was selected to indicate the shape indirectly of the interface between the molten aluminum and the electrolyte, as shown in Figure 20. The instant pressure difference at 10 seconds after gas was injected into the model was used in the profile of the metal–electrolyte. The interface with a 1.1 deg tilted anode is flatter than the interface with a 0 deg tilted anode.

IX. CONCLUSIONS

In this study, a full-scale water model and mathematical models were established to investigate the fluid flow related phenomena in an aluminum electrolysis cell. The following conclusions were derived:

1. The 3D time-dependent multiphase numerical model can predict the transient phenomena in the electrolysis bath. The accuracy of the model is proven by the ink dispersion and LDV water modeling measurement.
2. The morphology of the gas bubble under the anode is recorded by a video camera in the water model and was predicted by the 3D mathematical simulation. The formation of a gas film implied a high resistance and energy consumption.
3. The LDV measurements revealed a recirculation flow pattern in the side channel, similar to the observation of the ink dispersion. With the increasing of the anode-tilted angle, the flow velocity and turbulent energy increased in this region.
4. The design of the anode corner shape has a great effect on the bubble release. The 90 deg anode edge design delayed the bubble floatation. Bubbles tended to move forward at the anode rectangle edge, whereas the bubble could move along the curved anode corner edge and the detached point of bubble

was close to the anode. Therefore, the curved anode edge was more efficient for bubble removal than the 90 deg anode edge design.

5. The bubble release frequency was calculated by a 3D mathematical simulation. The bubble release frequency was pseudoperiodical. The 1.1 deg tilted-angle anode had a larger release frequency than that of the 0 deg tilted angle anode. Generally, the lower the bubble release frequency, the greater the bubble size under the anode.
6. The metal-electrolyte interface was predicted by correlating the pressure distribution to the height of the interface. The interface with a 1.1 deg tilted anode was more stable than that with a 0 deg tilted anode.
7. The anode design coupling the curved corner edge, with slot and with a tilted angle, is efficient for bubble releasing and for electrolyte-metal stability.

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REFERENCES

1. R.J. Aaberg, V. Ranum, K. Williamson, and B.J. Welch: *Light Metals*, TMS, Warrendale, PA, 1997, pp. 341–46.
2. J. Zoric and A. Solheim: *J. Appl. Electrochem.*, 2000, vol. 30, pp. 787–94.
3. T.M. Hyde and B.J. Welch: *Light Metals*, TMS, Warrendale, PA, 1997, pp. 333–40.
4. J. Xue and H. Øye: *Light Metals*, TMS, Warrendale, PA, 1995, pp. 265–70.
5. W.E. Haupin: *JOM*, 1971, vol. 23 (7), pp. 41–44.
6. H. Vogt: *Electrochim Acta*, 1997, vol. 42 (17), pp. 2695–705.
7. G. Mamantov and R. Marassi: *Molten Salt Chemistry: An Introduction and Selected Applications*, D. Reidel, Dordrecht, the Netherlands, 1987.
8. R. Shekhar and J.W. Evans: *Light Metals*, TMS, Warrendale, PA, 1990, pp. 243–48.
9. K.J. Fraser, M.P. Taylor, and A.M. Jenkin: *Light Metals*, TMS, Warrendale, PA, 1990, pp. 221–26.
10. Y. Wang and L. Zhang: in *Numerical Modeling on the Fluid Flow-Related Phenomena in an Aluminum Electrolysis Cell*, B.Q. Li, B.G. Thomas, L. Zhang, F.M. Doyle, and A.P. Campbell, eds., TMS, Warrendale, PA, 2010, pp. 207–15.
11. L. Zhang, Y. Wang, and X. Zuo: *Fluid Flow and Bubble Behavior in the Aluminum Electrolysis Cell*, Norwegian University of Science and Technology, Report, 2007.
12. G. Bearne, D. Gadd, and S. Lix: *Light Metals*, TMS, Warrendale, PA, 2007, pp. 305–10.
13. A. Panaitescu, A. Moraru, and I. Panaitescu: *Light Metals*, TMS, Warrendale, PA, 2003, pp. 359–66.
14. O. Kobbeltvedt and B.P. Moxnes: *Light Metals*, TMS, Warrendale, PA, 1997, pp. 369–76.
15. T. Utigard and J.M. Toguri: *Light Metals*, TMS, Warrendale, PA, 1986, pp. 405–11.
16. X. Wang and A.T. Tabereaux: *Light Metals*, TMS, Warrendale, PA, 2000, pp. 239–47.
17. W.E. Wahnsiedler: in *Light Metals*, R.D. Zabreznik, ed., TMS, Warrendale, PA, 1987, pp. 269–87.
18. S. Fortin, M. Gerhardt, and A.J. Gesing: *Light Metals*, TMS, Warrendale, PA, 1984, pp. 721–41.
19. X. Wang, G. Tarcy, S. Whelan, and S. Porto: in *Light Metals*, Morten Sorlie, ed., TMS, Warrendale, PA, 2007, pp. 299–304.
20. J.J.J. Chen, Z. Jianchao, Q. Kangxing, B.J. Welch, and M.P. Taylor: *The Fifth Asian Congress of Fluid Mechanics*, 1992, pp. 1173–76.
21. W. Yang and M.A. Cooksey: *Light Metals*, TMS, Warrendale, PA, 2007, pp. 451–56.
22. M.A. Cooksey and W. Yang: *Light Metals*, TMS, Warrendale, PA, 2006, pp. 359–65.
23. R. Shekhar and J.W. Evans: *Metall. Mater. Trans. B*, 1994, vol. 25B, pp. 333–40.
24. R. Shekhar and J.W. Evans: *Metall. Mater. Trans. B*, 1994, vol. 25B, pp. 341–49.
25. R. Shekhar and J.W. Evans: *Metall. Mater. Trans. B*, 1996, vol. 27B, pp. 19–27.
26. K.E. Einarsrud: *Metall. Mater. Trans. B*, 2010, vol. 41B, pp. 560–73.
27. S.D. Lympny, D.P. Ziegler, and J.W. Evans: *Light Metals*, TMS, Warrendale, PA, 1983, pp. 507–17.
28. M. Segatz: in *Light Metals 1993*, S.K. Das, ed., TMS, Warrendale, PA, 1993, pp. 355–60.
29. T. Haarberg, A. Solheim, S.T. Johansen, and P.A. Solli: *Light Metals*, TMS, Warrendale, PA, 1998.
30. L.I. Kiss: *Fifth International Conference on CFD in the Process Industries*, Melbourne, Australia, 2006, pp. 1–7.
31. M.M. Bilek, W.D. Zhang, and F.J. Stevens: *Light Metals*, TMS, Warrendale, PA, 1994, pp. 323–31.
32. M.A. Doheim, A.M. El-Kersh, and M.M. Ali: *Metall. Mater. Trans. B*, 2007, vol. 38B, pp. 113–19.
33. X. Li, J. Li, and Y. Lai: *Trans. Nonferrous Met. Soc. China*, 2004, vol. 14 (6), pp. 1221–26.
34. F.N. Ngoya and J. Thonstad: *Electrochim. Acta*, 1985, vol. 30, pp. 1659–64.
35. N. Zhou and X. Xia: *Light Metal*, 2004, vol. 12, pp. 26–30.
36. M.A. Cooksey and W. Yang: in *Light Metals*, T.J. Galloway, ed., TMS, Warrendale, PA, 2006, pp. 359–65.
37. L.I. Kiss, S. Poncsák, and J. Antille: *Light Metals*, TMS, Warrendale, PA, 2005, pp. 559–64.
38. L.I. Kiss, A.L. Perron, S. Poncsák, J. Antille, and T. Nguyen: in *Light Metals 2007*, M. Sorlie, ed., TMS, Warrendale, PA, 2007, pp. 495–99.
39. Fluent, Inc.: *FLUENT6.1-Manual*, Lebanon, NH, 2003.
40. B.G. Thomas, Q. Yuan, S. Sivaramakrishnan, T. Shi, S.P. Vanka, and M.B. Assar: *ISIJ Int.*, 2001, vol. 41 (10), pp. 1262–72.
41. B.E. Launder and D.B. Spalding: *Comp. Meth. Applied Mechanics Engr.*, 1974, vol. 13 (3), pp. 269–89.
42. B.G. Thomas and L. Zhang: *ISIJ Int.*, 2001, vol. 41 (10), pp. 1181–93.
43. D.R. Sadoway: *Light Metals*, TMS, Warrendale, PA, 1990, pp. 403–07.
44. D.F. Che, J.J.J. Chen, and M.P. Taylor: *Int. Conf on CFD in Mineral and Metal Processing and Power Generation*, Institution of Chemical Engineers, 1991, pp. 25–29.
45. B.P. Moxnes, B.E. Aga, and J.H. Skaar: *Light Metals*, TMS, Warrendale, PA, 1998, pp. 247–55.
46. Y. Feng, M.A. Cooksey, and P. Schwarz: *Light Metals*, TMS, Warrendale, PA, 2007, pp. 339–44.
47. B. Ren, Z. Wang, Z. Shi, Y. Ban, and Z. Qiu: *Mining Metall. Eng.*, 2007, vol. 27 (3), pp. 61–63.